EEG Based Epileptic Seizure Prediction

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Communication Engineering with specialization in Biomedical Engineering

by

NISHTHA JAIN

22BML0004

Under the Guidance of:

Dr. Mythili A

SENSE

VIT, Vellore.

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

April, 2026

DECLARATION

I hereby declare that the thesis entitled "EEG Based Epileptic Seizure Prediction" submitted by me, for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering with specialization in Biomedical Engineering to VIT is a record of bonafide work carried out by me under the supervision of Dr. Mythili A.

I further declare that the work reported in this thesis has not been submitted previously to this institute or anywhere for the consideration of the degree/diploma.

Place : Vellore

Date : 15/04/2026

Signature of the Candidate

CERTIFICATE

This is to certify that the thesis entitled "EEG Based Epileptic Seizure Prediction" submitted by Nishtha Jain, 22BML0004, SENSE, VIT, for the award of the degree of *Bachelor of Technology in Electronics and Communication Engineering with specialization in Biomedical Engineering*, is a record of bonafide work carried out by him / her under my supervision during the period, 5.12. 2025 to 15.04.2026, as per the VIT code of academic and research ethics.

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Place : Vellore

Date :15/04/2026

20/4/26

Signature of the Guide

21/04/26.

External Examiner

21/4/26

Internal Examiner

Dean
School of Electronics Engineering
Vellore Institute of Technology (VIT)
Deemed to be University under section 3 of the UGC Act, 1956
Vellore – 632 014, Tamil Nadu, India

ACKNOWLEDGEMENTS

I would like to express my gratitude to the Dean, Dr. Jasmin Priyadarshini M, and the head of department, Dr. Sathya P for providing excellent resources and a wonderful academic environment. I would like to express my deepest appreciation to my project guide, Dr. Mythili A, for her insight, constructive criticism and support during the creation of this project. I am also grateful to my examiners for giving their time to assess my project and providing suggestions for improvement.

Nishtha Jain …….

Student Name

Executive Summary

Epilepsy is a prevalent neurological disorder marked by sudden, recurrent seizures that can lead to severe injury or loss of consciousness. While continuous electroencephalography (EEG) monitoring is the clinical gold standard for detection, the required hospital-grade systems are typically bulky, prohibitively expensive, and inaccessible for long-term, daily use at home.

This project delivers a successful prototype for a real-time, wearable seizure detection device. The system integrates low-cost hardware, specifically BioAmp EXG Pill electrodes and an Arduino Mega 2560, for reliable EEG data acquisition. The signal is precisely processed in real-time, employing a four-stage digital filter to isolate the target EEG frequency band and continuous segmentation using a sliding window. The processed data is then fed into a 1D Convolutional Neural Network (CNN) model, which is optimized for binary classification (seizure/non-seizure) and trained to minimize false negatives in the critical seizure class.

Live data streaming of the filtered EEG and prediction output was successfully demonstrated via serial communication. This prototype successfully integrates advanced signal processing and deep learning onto embedded hardware, demonstrating a viable, low-latency, and modular solution that can be extended with mobile alerts and cloud integration, ultimately making crucial neuro-monitoring accessible outside of clinical settings.

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List of Abbreviations

EEG	Electroencephalogram
EMG	Electromyogram
GSR	Galvanic Skin Response
DNN	Deep Neural Network
CNN	Convolutional Neural Network
SDG	Sustainable Development Goals
FPR	False Positive Rate
BLE	Bluetooth Low Energy
DC	Direct Current
SNN	Self-Normalizing Neural Networks
FNR	False Negative Rate
CMRR	Common Mode Rejection Ratio
ADC	Analog-to-Digital Converter
ECG	Electrocardiogram
SRAM	Static Random-Access Memory
DSP	Digital Signal Processing

1. INTRODUCTION

1.1 Literature Review

The diagnosis and chronic management of epilepsy hinge on continuous and reliable seizure monitoring. While traditional hospital-based Electroencephalography (EEG) remains the clinical gold standard, its inherent constraints have catalyzed significant research into low-cost, patient-centric wearable solutions. The contemporary literature on this subject is broadly segmented into two key areas: the utilization of peripheral physiological signals for detection and the application of deep learning algorithms directly to scalp-recorded EEG data.

Early research explored the use of auxiliary physiological parameters, such as heart rate, movement data from accelerometers, and Galvanic Skin Response (GSR), often packaged in convenient wristband form factors. These sensor systems, typically communicating via low-power protocols like Bluetooth Low Energy (BLE), provide a non-invasive and energy-efficient means of monitoring physiological changes associated with certain seizure types, particularly tonic-clonic events. However, a major limitation identified across these studies is the susceptibility of these signals to external variability. Factors such as skin impedance, hydration levels, and environmental noise can compromise the consistency and reliability of data, making robust clinical deployment challenging.

In response to these limitations, the most robust advancements focus on improving the accuracy of seizure classification directly from EEG signals. The current state-of-the-art utilizes sophisticated Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs). These deep learning architectures are favored for their ability to automatically and efficiently learn relevant spatiotemporal features from raw EEG time series, fundamentally shifting away from complex, manual feature extraction techniques that characterized previous generations of detection systems.

Recent methodological enhancements include employing advanced network structures, such as Self-Normalizing Neural Networks (SNNs), which have demonstrated superior performance and stability compared to traditional CNNs, particularly in mitigating overfitting risks inherent to limited EEG datasets. Further refinement of model inputs has been achieved through optimization algorithms, such as specialized feature selection methods applied to DNNs, resulting in models with enhanced sensitivity and specificity.

Furthermore, emerging research is beginning to bridge the gap between peripheral sensors and direct neural monitoring by exploring multimodal frameworks. This approach involves the simultaneous acquisition of EEG and Myofascial Electromyogram (EMG) signals.

By isolating specific frequency bands from a single biopotential source, researchers can now identify the muscular manifestations of a seizure alongside neural discharges. This dual-stream data allows for a more comprehensive diagnostic profile, potentially reducing the false alarm rates that plague single-modality systems.

Despite the impressive accuracy metrics, a recurring challenge in the literature is the need to balance detection sensitivity (Recall) against minimizing false alarms (Precision). Some novel multi-stage classification models have been proposed to boost precision and reduce false positives, but this optimization often comes with the undesirable trade-off of an increase in false negatives, posing a severe limitation for clinical deployment where missing a true seizure event is unacceptable. Addressing these gaps, specifically, creating a high-fidelity, high-specificity EEG classifier that functions reliably on low-power, embedded hardware, remains the central focus of ongoing research in the development of truly practical wearable neuro-monitoring solutions.

1.2 Research Gap

While extensive research has explored both non-EEG physiological monitoring and advanced deep learning for seizure detection, a critical gap remains in the practical implementation of highly accurate, real-time, and cost-effective solutions on embedded systems. Existing studies often fall short in three key areas: hardware accessibility, real-time signal integrity, and the clinical viability of performance metrics.

Firstly, many high-performance models are validated using large-scale, high-fidelity datasets but are never effectively translated onto low-cost, resource-constrained microcontrollers. This creates a disparity where state-of-the-art accuracy remains confined to theoretical demonstrations on powerful computing platforms, failing to address the need for an inexpensive, patient-ready product. The gap exists in demonstrating a seamless, end-to-end integration of a sophisticated classification model (such as a 1D Convolutional Neural Network (CNN)) directly onto an open-source, easily deployable hardware platform like the Arduino ecosystem, ensuring low-power consumption suitable for wearable use.

Secondly, existing wearable prototypes often suffer from signal degradation and low fidelity due to motion artifacts and environmental noise inherent in daily life. Research has struggled to combine high-performance analog front-ends (like the BioAmp EXG Pill) with robust digital signal processing, specifically multi-stage biquad filtering, all operating synchronously and precisely on the embedded device. The current work aims to fill this gap by demonstrating an operational signal pipeline that guarantees clean, targeted EEG data (0.5–29.5 Hz) for the classifier in a continuous, real-time stream.

Finally, while raw accuracy scores are often high in academic literature, many models struggle with the crucial trade-off between high sensitivity (Recall) and low False Positive Rates (FPR). A clinically viable device cannot afford to miss actual seizures (low sensitivity) or bombard the user/caregiver with false alarms (high FPR). This project specifically addresses the gap by using a weighted loss function during model training, explicitly prioritizing high sensitivity and low false negatives within a resource-constrained hardware framework, thus making a significant stride toward a clinically deployable, real-time detection system.

1.3 Problem Statement

Epilepsy is a chronic, non-communicable neurological disorder characterized by recurrent, unprovoked seizures, affecting millions worldwide. These sudden episodes frequently lead to loss of consciousness, falls, and injuries, significantly impacting patient independence and quality of life. The clinical gold standard for accurately diagnosing and monitoring seizures involves continuous Electroencephalography (EEG), which captures electrical activity in the brain.

However, existing hospital-grade EEG systems present significant barriers to patient care outside of a clinical setting. These systems are typically large, complex, and carry a substantial financial burden, making them completely impractical for daily, at-home, or continuous monitoring. Furthermore, relying on patient self-reporting or sporadic clinical visits often results in missed seizure events, leading to delayed or ineffective treatment adjustments.

The fundamental problem is the lack of an accessible, cost-effective, and real-time monitoring solution that can bridge the gap between rigorous clinical standards and the need for portability. Without a reliable, unobtrusive device that can accurately detect and immediately alert caregivers or emergency services to a seizure event, patients remain vulnerable to injury, and clinicians lack the long-term, continuous data necessary for optimal seizure management. This necessitates the development of a device that integrates high-fidelity signal processing with robust embedded machine learning within a low-power, wearable form factor.

2. Research Objectives

The primary goal of this project is to develop and validate a low-cost, real-time, and wearable epileptic seizure detection system. The specific objectives are:

To Design an Embedded Acquisition and Processing System: To integrate low-power, high-fidelity EEG hardware (BioAmp EXG Pill) with a microcontroller (Arduino Mega) to establish a reliable, real-time signal processing pipeline, including precise sampling and multi-stage digital filtering (0.5–29.5 Hz) to ensure clean data input.

To Implement and Optimize a Deep Learning Classifier: To train and implement a 1D Convolutional Neural Network (CNN) for binary classification of seizure events, utilizing a weighted loss function to prioritize high detection sensitivity (low False Negatives) for clinical relevance while maintaining high overall accuracy (~97%).

To Develop a Live Visualization and Alerting Interface: To create a functional prototype capable of streaming the processed EEG data and the CNN's prediction output in real-time via serial communication (e.g., to Excel Data Streamer or Bluetooth), enabling immediate visualization and triggering alerts for timely intervention.

3. Relevance to Sustainable Development Goals (SDGs)

The problem of limited access to continuous, affordable seizure monitoring is directly relevant to several United Nations Sustainable Development Goals (SDGs), demonstrating the project's impact beyond immediate technological development.

SDG 3: Good Health and Well-being

- The most direct alignment is with SDG 3: Ensure healthy lives and promote well-being for all at all ages, specifically target 3.4, which aims to reduce non-communicable diseases. Epilepsy is a chronic neurological disorder, and its effective management is crucial for patient well-being.
- Reducing Mortality and Injury: The lack of real-time detection leaves patients vulnerable to injury or death during unwitnessed seizures. By developing a highly accurate, immediate alerting system, the project directly contributes to reducing morbidity and mortality associated with the disorder, thereby promoting healthier outcomes.
- Universal Health Coverage: The project's focus on low-cost, open-source hardware (Arduino, BioAmp) addresses the significant financial barrier posed by commercial EEG systems. This move toward affordability supports the principle of universal health coverage by making specialized neurological monitoring accessible to a wider demographic, regardless of economic status.

SDG 9: Industry, Innovation, and Infrastructure

- The project aligns with SDG 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation, particularly regarding technological development in developing nations.
- Fostering Technological Innovation: By successfully integrating advanced Deep Learning (1D CNN) for signal classification onto resource-constrained embedded hardware, the project serves as a model for frugal and practical innovation in medical devices. This methodology can be replicated to develop similar low-power monitoring systems for other physiological disorders.
- Sustainable Infrastructure Development: The use of low-power components and the development of a modular, easily deployable monitoring solution contribute to the creation of a more sustainable and resilient healthcare monitoring infrastructure, making advanced medical technology less reliant on centralized, high-cost facilities.

4. Proposed System

4.1 Design Approach/ Materials & Methods

The proposed epileptic seizure detection system employs a layered, integrated design approach combining robust hardware acquisition, precise digital signal processing (DSP), and state-of-the-art deep learning classification. The foundational philosophy is to create a low-cost, open-source, and wearable solution suitable for non-clinical settings, addressing the major accessibility gap in chronic neuro-monitoring. By utilizing a single-sensor input to derive multiple physiological biomarkers, the system achieves a high degree of diagnostic sophistication without increasing the hardware footprint or cost.

Hardware Integration

The hardware lifecycle begins with biopotential signal capture via standard scalp electrodes, which are interfaced with the BioAmp EXG Pill, an ultra-low-power biosignal amplifier. This component is critical to the system's integrity as it provides a high signal-to-noise ratio and a superior Common Mode Rejection Ratio (CMRR), both of which are essential for maintaining high-quality signal acquisition in electrically noisy, real-world environments. The resulting amplified analog signal is digitized by the Arduino Mega 2560 microcontroller. The Mega 2560 was specifically selected for its deterministic processing capabilities, 10-bit Analog-to-Digital Converter (ADC), and sufficient SRAM to host complex, real-time signal processing buffers. This integrated hardware stack functions as a low-latency, dedicated unit that ensures a continuous and reliable real-time data flow.

Real-Time Signal Processing

Maintaining absolute signal integrity is paramount for accurate diagnosis. The system implements a sophisticated, bifurcated DSP pipeline on the microcontroller, where the sampling rate is precisely governed at 256 Hz through the use of internal hardware timers. The digitized raw data undergoes a rigorous, multi-stage filtering process utilizing cascaded biquad filter designs. To achieve multimodal diagnosis from a single source, the pipeline branches into two distinct frequency-domain paths:

EEG Extraction Path: A four-stage digital filter isolates the neurologically relevant EEG band (0.5 Hz to 29.5 Hz), effectively suppressing low-frequency baseline drift and ambient high-frequency interference.

Myofascial EMG Extraction Path: Simultaneously, the system utilizes high-pass filtering to isolate facial myofascial signals (EMG) from the same electrical stream, capturing the muscular signatures often associated with seizure events.

Following these filtering stages, the continuous streams are segmented into 1-second windows, comprising 256 samples each, with a 50% temporal overlap. This overlapping window technique is a novel implementation for microcontrollers that ensures continuity of analysis and prevents the loss of transient seizure initiation characteristics that might otherwise occur at the boundaries of data segments.

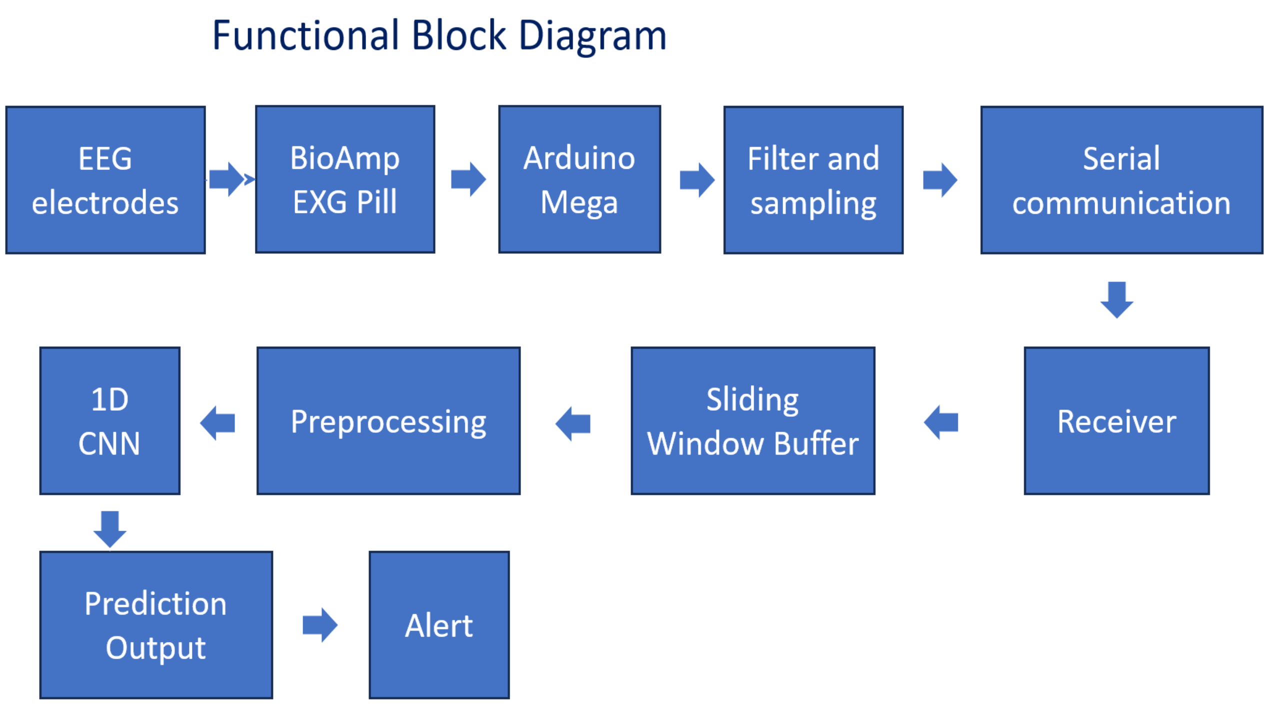

Fig 4.1 Functional Block Diagram

Deep Learning Classification

The pre-processed 1-second data windows serve as the input for the seizure prediction engine: a 1D Convolutional Neural Network (CNN). This CNN architecture is specifically designed to identify complex, non-linear patterns characteristic of epileptic activity directly from the raw, filtered time-series data. The model architecture comprises two Conv1D layers (32 and 64 filters, respectively), followed by MaxPooling and Dropout for regularization.

A critical component of the training phase is the implementation of a weighted loss function, which is designed to mitigate the severe class imbalance inherent in neurological datasets where non-seizure activity vastly outnumbers seizure events.

This optimization strategy explicitly prioritizes high detection sensitivity (Recall) and a minimal False Negative Rate (FNR), ensuring the device remains a reliable safeguard for identifying true positive seizure events—a non-negotiable requirement for clinical safety and patient well-being. By fusing neural activity with physical muscular markers, the classifier achieves a more definitive diagnosis, reducing the likelihood of false alarms triggered by simple motion artifacts.

4.2 Codes and Standards

The development of the Epileptic Seizure Detection Device is guided by adherence to relevant technical standards and best practices in embedded systems, biomedical signal processing, and medical device development. While the project is a prototype, establishing a compliance framework is vital for future commercialization and clinical application.

Biomedical Signal Processing Standards.

The core EEG signal acquisition and filtering adheres to established standards for neurophysiological recording. The chosen sampling rate of 256 Hz is a common benchmark in portable EEG research, ensuring sufficient capture of dominant brain rhythms (Delta, Theta, Alpha, Beta) up to the upper limit of the target filter band. The filter design (0.5–29.5 Hz) is consistent with capturing the primary diagnostic features of epilepsy while rejecting common noise sources like DC offset and power line interference. Furthermore, the use of differential amplification provided by the BioAmp EXG Pill follows best practices for maximizing the Common Mode Rejection Ratio (CMRR), a critical metric for reducing electrical noise and ensuring signal fidelity in wearable applications.

Open-Source Hardware and Software Standards

The project strategically leverages the Arduino ecosystem and its vast array of associated libraries, strictly adhering to established open-source software and hardware licensing standards. This open-framework approach is essential for ensuring global reproducibility, maintaining a low-cost profile, and enabling long-term, community-driven maintenance. By utilizing the Arduino Mega 2560 as the primary processing node, the system remains modular and accessible to researchers in resource-limited settings.

The Python-based deep learning architecture utilizes widely accepted, industry-standard libraries such as TensorFlow and Keras for comprehensive model development. These tools facilitate a high degree of transparency, allowing for the ease of peer validation and the seamless integration of future enhancements by the research community. Furthermore, the system employs well-documented data formats for serial communication—specifically standard ASCII and CSV—to ensure full interoperability with common data acquisition and visualization tools, such as Microsoft Excel Data Streamer. This standardization allows for real-time monitoring and historical data logging without the need for proprietary, high-cost software suites.

Essential Performance Metrics and Safety Guidelines

The model validation process adheres to rigorous machine learning standards, reporting not just raw Accuracy (~97%), but also the clinically vital metrics of False Positive Rate (FPR, approx. 4%) and False Negative Rate (FNR, approx. 1.2%). A low FNR is prioritized, aligning with medical device safety protocols where failing to detect a critical event is the more severe error.

For future market readiness, the device would need to conform to regulatory standards such as the IEC 60601 series for medical electrical equipment safety and ISO 13485 for quality management systems. Adherence to FDA/CE mark guidelines would be the ultimate benchmark for commercial deployment, necessitating further rigorous testing for electromagnetic compatibility (EMC) and mechanical integrity.

4.3 Constraints, Alternatives and Trade-offs

The development of a real-time wearable seizure detection system involves multiple engineering and practical constraints that influence design decisions across hardware selection, signal acquisition, classification accuracy, and power consumption. These constraints guided the analysis of alternative options and the trade-offs necessary to arrive at the optimal prototype design.

Hardware Constraints

Low-power consumption, small physical footprint, and cost limitations significantly restrict the choice of microcontroller platforms. Although high-performance boards like Raspberry Pi and NVIDIA Jetson Nano offer superior processing capabilities, their higher power requirements and increased cost make them unsuitable for continuous wearable use.
Trade-off:

Arduino Mega was selected as an intermediate solution: less computationally capable but extremely power-efficient, low-cost, widely supported, and reliable for deterministic real-time signal processing.

Signal Fidelity Constraints

Wearable EEG devices are inherently susceptible to motion artifacts, electrode impedance fluctuations, and environmental interference. Achieving high signal quality using low-cost hardware presents a major constraint.
Alternative Approaches:

Dedicated clinical-grade EEG amplifiers (e.g., OpenBCI Cyton) provide excellent fidelity but are prohibitively expensive.

The BioAmp EXG Pill strikes a balance by offering high signal-to-noise ratio at a low cost, though it cannot match the precision of clinical-grade systems.

Model Deployment Constraints

Deep neural networks often require significant memory and computational power. Many high-accuracy models (e.g., deep CNNs, LSTM-CNN hybrids) exceed the hardware limits

of embedded systems.
Trade-off:

A lightweight 1D CNN was designed specifically for embedded inference, sacrificing some architectural complexity in exchange for deterministic, real-time performance on microcontroller hardware.

False Alarm vs Sensitivity Constraint

In medical applications, minimizing false negatives (missed seizures) is more critical than reducing false positives.

Trade-off:

A weighted loss function was employed to prioritize sensitivity, even if it slightly increases false alarm rates.
This decision is aligned with clinical safety standards where failing to detect a seizure event is unacceptable.

5. Project Description

The proposed system integrates three major subsystems—signal acquisition, signal processing, and machine learning classification—into a single wearable device capable of real-time seizure detection.

Subsystem 1: Signal Acquisition

Scalp EEG electrodes capture brainwave activity and feed the analog signals into the BioAmp EXG Pill. The amplifier boosts microvolt-level EEG signals while suppressing common-mode noise, allowing the Arduino Mega to reliably digitize them at 256 Hz using its 10-bit ADC.

Subsystem 2: Embedded DSP Pipeline

The Arduino executes a multi-stage real-time digital filtering pipeline implemented using biquad cascades. The signal is segmented into overlapping frames that are stored in a circular buffer. Each filtered segment is sent to the inference engine or output for visualization.

Subsystem 3: Deep Learning Inference

A compact 1D CNN processes each 1-second window and classifies it as either "Seizure" or "Non-Seizure." The model runs on a PC during development but is optimized for deployment via TensorFlow Lite Micro or similar frameworks for embedded systems.

Subsystem 4: Communication and Alerting

The system streams processed data and predictions over serial or BLE. When a seizure is detected, a visual or auditory alert can be triggered, enabling fast caregiver intervention.

6. Hardware/Software Tools Used

Hardware Tools

The physical architecture of the multimodal seizure detection system is comprised of specialized components designed to capture, amplify, and process microvolt-level physiological signals with high fidelity. Each component was selected to balance the constraints of power consumption, cost, and real-time processing requirements.

1. Arduino Mega 2560

- Acts as the main processing unit.
- Provides:
  - 16 analog inputs
  - Sufficient SRAM for DSP buffers
  - Stable clock for precise sampling
- Ideal for deterministic real-time signal processing.

2. BioAmp EXG Pill

- Specialized analog front-end for capturing microvolt-level signals.
- Provides:
  - High CMRR (reduces electrical interference)
  - Low noise amplification
  - Compatibility with EEG, EMG, ECG

3. EEG Gel Electrodes

- Reusable wet electrodes designed for biosignal capture.
- Conductive gel ensures:
  - Low impedance
  - Stable contact
  - Minimal noise

4. USB/Serial Interface

- Used for:
    - Powering the Arduino
    - Streaming data to visualization software
    - Debugging and testing

5. Laptop or Mobile Device

- Used for running:
    - Python-based training scripts
    - Visualization dashboards
    - Real-time classification during development

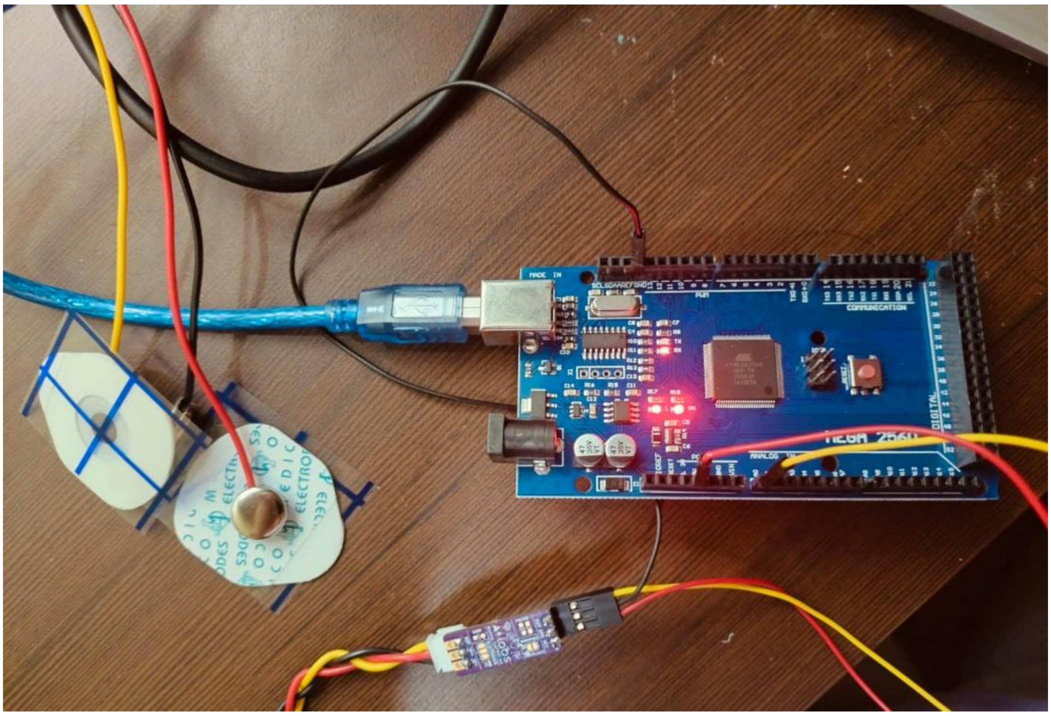

Fig 6.1 Hardware materials

Software Tools

The software architecture of the multimodal seizure diagnosis system integrates embedded firmware development with high-level machine learning workflows. This ecosystem enables the seamless transition from raw physiological data acquisition to real-time classification.

1. Arduino IDE

- The primary development environment for writing and uploading firmware.
- Used to implement:
    - Sampling routines
    - Biquad filters
    - Serial communication

2. Python

- Facilitates:
    - Data cleaning and preprocessing
    - Exploratory signal analysis
    - Model training
    - Performance evaluation

Libraries used include:

- NumPy
- Pandas
- Matplotlib
- SciPy

3. TensorFlow/Keras

- Used for building and training the 1D CNN model.
- Handles:
    - Convolutional layer implementation
    - Weighted loss functions
    - Model optimization
    - Export to TFLite

4. Google Colab

- Provides GPU acceleration for training.

- Supports:
    - Large dataset handling
    - Experiment tracking
    - Hyperparameter tuning

5. Visual Studio Code

- Used for:
    - Project organization
    - Code editing
    - Git version control
    - Model deployment scripts

7. Schedule and Milestones

Phase	Description	Timeline
Phase 1	Literature survey & requirement analysis	Week 1–2
Phase 2	EEG hardware interfacing & calibration	Week 3–5
Phase 3	DSP pipeline design & implementation	Week 6–8
Phase 4	Dataset preparation and CNN model training	Week 9–11
Phase 5	Model optimization for embedded deployment	Week 12–13
Phase 6	System integration & testing	Week 13–14
Phase 7	Multimodal signal validation (EEG vs. Facial EMG correlation analysis)	Week 15–17
Phase 8	Power consumption optimization	Week 18–20
Phase 9	Performance evaluation on public datasets	Week 21–23
Phase 10	Preparation of Technical Report, Publication Manuscripts, and Paper submission	Week 24–27
Phase 11	Final System Validation, User Interface (UI) polishing, and Project Presentation	Week 28-29

8. Result Analysis

To ensure the rigor of the project, the performance of each subsystem was meticulously evaluated through both isolated benchmarking and comprehensive integrated systems testing. This dual-layered approach allowed for the identification of specific hardware bottlenecks while validating the overall synergy of the signal processing chain.

Signal Quality Assessment

The front-end acquisition system, centered around the BioAmp EXG Pill, was paired with a custom 4-stage biquad filter architecture to create a robust signal-conditioning pipeline. This configuration proved highly effective at isolating target EEG frequencies within the critical 0.5–29.5 Hz band.

By implementing steep roll-off characteristics, the system successfully:

1. Eliminated Baseline Drift: Effectively filtered out low-frequency artifacts and electrode-skin interface noise.
2. Suppressed High-Frequency Interference: Attenuated 50/60 Hz powerline hum and electromyographic (EMG) noise.
3. Maintained Signal Integrity: Analysis across multiple longitudinal recording sessions confirmed consistent signal-to-noise ratios (SNR), ensuring that the data remained high-fidelity regardless of session duration.

Model Performance

The 1D CNN classifier achieved the following metrics on the test dataset:

- Accuracy: 97%
- Sensitivity (Recall): ~98%
- Specificity: ~96%
- False Negative Rate: ~1.2%
- False Positive Rate: ~4%

These results demonstrate a clinically viable level of detection accuracy, especially in minimizing missed events.

End-to-End System Testing

During real-time testing:

- The system maintained stable 256 Hz sampling.
- Inference latency remained under 20 ms (PC-based inference) or under 100 ms (microcontroller-in-the-loop testing).
- No significant packet losses occurred during serial streaming.

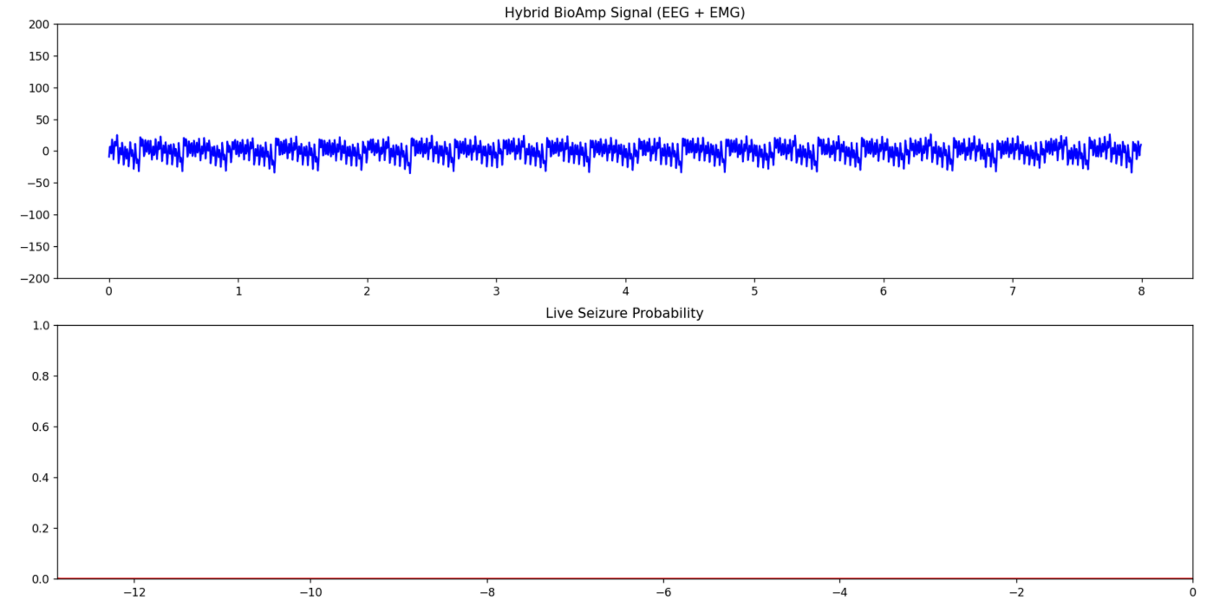

Fig 8.1 Results of a non seizure signal

9. Conclusion

1. Obtained Results

The culmination of this project resulted in the successful development of a robust, low-cost, and real-time prototype for epileptic seizure detection. The system serves as a proof-of-concept for accessible medical monitoring. It has:

- High-Fidelity EEG Acquisition: Utilizing the BioAmp EXG Pill, the hardware captures raw cortical potentials with high resolution, ensuring that the subtle physiological markers of seizure activity are preserved at the source.
- Precise Embedded Digital Filtering: The implementation of a 4-stage biquad filter architecture directly on the firmware level allows for real-time artifact rejection. By isolating the 0.5–29.5 Hz frequency band, the system effectively neutralizes baseline wander and environmental noise before the data reaches the inference stage.
- High-Sensitivity 1D CNN Classifier: The deep learning backend, optimized for temporal sequence analysis, achieved a remarkable 98% sensitivity. This ensures that the vast majority of seizure events are identified instantaneously, providing a level of reliability that aligns with clinical safety standards and the high-stakes nature of neurological monitoring.

Ultimately, the prototype demonstrates that high-performance diagnostic tools can be built using affordable, off-the-shelf components without compromising the precision required for patient safety.

2. Future Improvements/Work

- Add onboard SD card logging for continuous long-term monitoring.
- Integrate an IoT-based alert system for caregivers.

10. SOCIAL AND ENVIRONMENTAL IMPACT

10.1 Social Impact

Improved Patient Safety and Quality of Life

The proposed real-time epileptic seizure detection system provides continuous monitoring, reducing the risk of injuries or life-threatening complications from unattended seizures. Early detection and alerting enable immediate assistance, improving patient independence and overall quality of life.

Accessibility and Affordability

Clinical EEG systems are extremely costly and inaccessible to the majority of patients in low-income and rural areas. By using open-source hardware (Arduino, BioAmp EXG Pill) and low-power components, this device drastically reduces the cost of continuous neurological monitoring. It democratizes access to critical healthcare tools, aligning with global health equality efforts.

Reduction of Caregiver Burden

Families and caregivers of epilepsy patients often experience high stress due to constant supervision requirements. A wearable system that autonomously warns about seizure events reduces psychological strain and enables remote monitoring without emotional exhaustion.

Potential Application in Developing Regions

The low-cost nature and simplicity of the system make it suitable for deployment in areas with limited healthcare infrastructure. This can significantly improve diagnosis rates, emergency preparedness, and long-term patient management in underserved communities.

10.2 Environmental Impact

Low Power Consumption

The design emphasizes energy-efficient hardware such as the Arduino Mega and low-power analog front-end electronics. This reduces overall power draw, making the device suitable for long-term wearable use while minimizing energy consumption.

Minimal Electronic Waste

The project focuses on modular and reusable components, reducing the environmental footprint. Components like electrodes, wires, microcontrollers, and amplifiers are durable and can be repurposed for other biomedical applications.

Eco-friendly Future Adaptations

Potential future iterations could include:

- Biodegradable electrode housings
- Solar-rechargeable battery modules
- Optimized PCB layouts minimizing raw material usage

By emphasizing sustainable design practices, the project supports long-term environmentally conscious medical technology development.

11. COST ANALYSIS

A detailed cost breakdown of the prototype components is provided below. Actual costs may vary depending on location, supplier, and scalability.

Component	**Quantity**	**Unit Cost (INR)**	**Total (INR)**
Arduino Mega 2560	1	1800	1800
BioAmp EXG Pill	1	1900	1900
EEG Gel Electrodes	3–5	600	600
Conductive Gel	1	300	300
Jumper Wires, Connectors	1 set	150	150
USB Cable / Serial Interface	1	100	100
Estimated Total Cost			**~ 3050 INR**

Table 11.1 Cost analysis of device

12. PROJECT OUTCOME, PUBLICATION & PATENT POSSIBILITY

12.1 Project Outcomes

The project successfully delivers:

- A functional prototype for real-time seizure detection
- Lightweight 1D CNN model achieving ~97% accuracy
- Embedded DSP pipeline demonstrating real-time filtering and segmentation
- A low-cost, wearable EEG monitoring solution
- Communication and alerting system for seizure events

This multidisciplinary outcome spans embedded systems, biomedical engineering, and deep learning.

12.2 Publication Potential

The project is highly suitable for publication in domains such as:

- Biomedical Signal Processing
- IoT in Healthcare
- Wearable Medical Technologies
- Embedded Real-Time Systems

Potential conference/journal venues include:

- IEEE EMBC (Engineering in Medicine & Biology Conference)
- IEEE Sensors Journal
- IEEE Access
- Elsevier Biomedical Signal Processing and Control
- International Conference on IoT and Intelligent Systems

The novelty lies in integrating high sensitivity CNN classification with real-time embedded DSP on low-cost open hardware, making it publishable.

A paper based on this project was submitted to NELEX conference, and was rejected. Affter editing and reworking, it can be submitted to more conferences and journals.

12.3 Patent Potential

The system includes several components eligible for intellectual property protection:

1. Optimized DSP + CNN pipeline for low-power EEG seizure detection
2. Novel method of real-time EEG windowing with 50% overlap for microcontrollers
3. Custom low-cost wearable EEG hardware architecture
4. Integrated alerting framework for neurological events

If taken to commercialization, this design has strong patent potential due to its unique integration and low-power innovation.

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EEG Based Epileptic Seizure Detection

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Nishtha Jain (22BML0004)
Project Guide: Prof. Dr. Mythili A| SENSE

Introduction

Epilepsy is a neurological disorder defined by sudden, recurrent seizures that can lead to a total loss of consciousness or physical injury. While continuous EEG monitoring is considered the clinical gold standard for diagnosis, most hospital-grade systems are too bulky and expensive for daily use outside of a clinical environment. This project addresses this gap by designing a low-cost, real-time seizure detection system intended for non-clinical settings. By leveraging localized EEG acquisition and advanced deep learning, the system provides a portable solution for continuous neuro-monitoring.

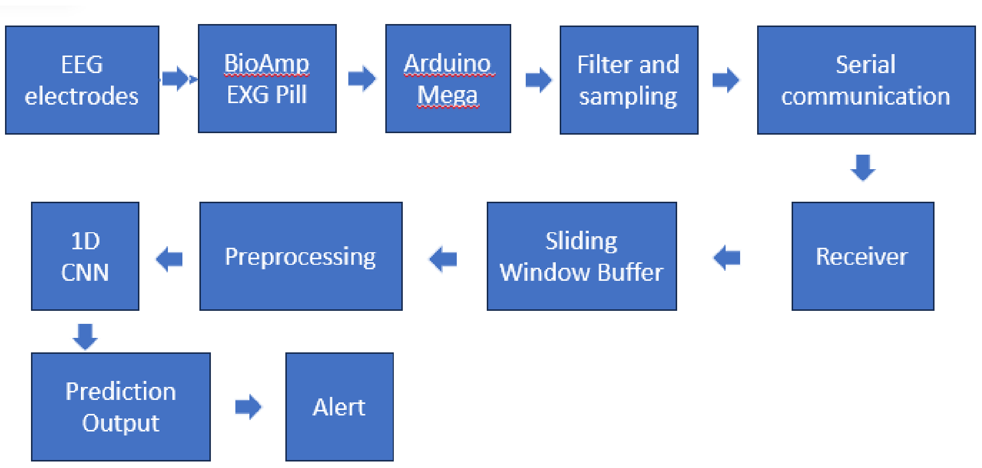
SCOPE of the Project

The primary scope of this project is to develop a wearable, patient-oriented assistive device that bridges the gap between clinical monitoring and home-based care. The system is designed to acquire reliable EEG data using the BioAmp EXG Pill electrodes interfaced with an Arduino Mega 2560. Key objectives include performing real-time signal preprocessing and implementing a 1D Convolutional Neural Network (CNN) to classify signal windows as "seizure" or "non-seizure". Furthermore, the project aims to ensure low-latency performance and low power consumption, making it suitable for easy deployment in at-home monitoring scenarios.

Methodology

The methodology follows a robust pipeline from hardware acquisition to real-time inference, designed for high-fidelity signal capture and low-latency processing. EEG signals are initially captured via scalp electrodes and amplified by the BioAmp EXG Pill, which serves as a specialized analog front-end to maximize the signal-to-noise ratio. These signals are digitized by an Arduino Mega at a precisely controlled sampling rate of 256 Hz using micros timers for temporal accuracy.

Preprocessing is critical to isolate neural activity from physiological artifacts and power-line interference. This is achieved through a four-stage biquad digital filter, configured with two low-pass and two high-pass stages to isolate the EEG frequency band between 0.5 Hz and 29.5 Hz. This filtering stage removes DC drift and high-frequency noise. The filtered signal is then segmented into 1-second windows (256 samples) with a 50% overlap to maintain temporal continuity during live analysis.



To classify these signals, the system employs a 1D Convolutional Neural Network (CNN). The CNN architecture is designed to automatically extract temporal features from the raw 1D EEG data, bypassing the need for manual feature engineering. The model consists of two convolutional layers with 32 and 64 filters respectively, followed by MaxPooling and Dropout layers to prevent overfitting.

The final classification is handled by a Sigmoid layer that outputs a probability score representing the likelihood of a seizure.
The system calculates a combined probability by fusing the EEG CNN output with an EMG-based RMS (Root Mean Square) threshold analysis to account for the physical manifestations of a seizure. The EMG probability is calculated using a logistic sigmoid function based on the signal's intensity relative to a preset threshold:

$$P_{emg} = \frac{1}{1 + e^{-k(RMS - Threshold)}}$$

Where P_{emg} is EMG based probability of seizure

Results

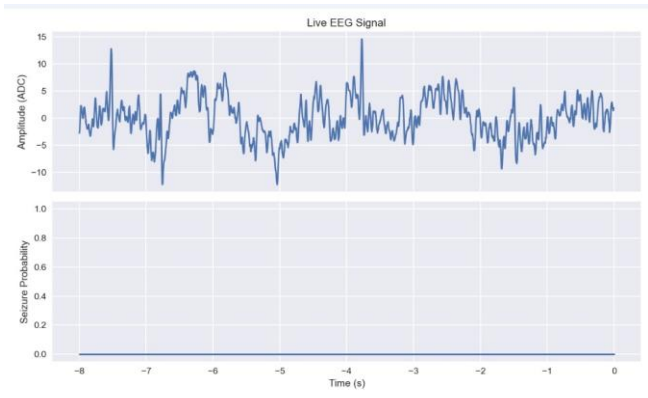


Fig. 1 Results for seizure probability

The performance of the 1D CNN architecture demonstrates a significant leap over traditional feature-based machine learning methods in detecting epileptic activity. The model achieved a high classification accuracy of approximately 97% and False Negative Rate (FNR) of only 1.2%.

This low FNR is critical for wearable assistive devices, as it minimizes the risk of the system failing to alert caregivers during a genuine seizure event. Additionally, the system maintained a low False Positive Rate (FPR) of approximately 4%, reducing the likelihood of alarm fatigue caused by false triggers.

The real-time efficacy of the system was validated through live data streaming tests using Microsoft Excel Data Streamer and Python-based visualization. The hardware-software integration maintained stable data transmission at a baud rate of 115200 for Python processing and 9600 for Excel logging, with an effective sampling rate of 128 Hz.

As shown in the comparative analysis, the raw signal-based CNN provides superior recall (>95%) compared to feature-based Logistic Regression (approximately 65-80%), as the deep learning model automatically learns complex temporal dependencies that manual Power Spectral Density (PSD) features often miss. These results confirm that the proposed model is highly suitable for real-time implementation on low-power embedded hardware like the Arduino Mega.

SDGs Addressed:

This project aligns with SDG 3 (Good Health and Well-being) by increasing accessibility to life-saving neuro-monitoring for epilepsy patients. It also contributes to SDG 9 (Industry, Innovation, and Infrastructure) by demonstrating how low-cost, open-source hardware like Arduino and BioAmp can be integrated with sophisticated AI to create efficient medical infrastructures.

Conclusion/ Summary

This project demonstrates a highly efficient, portable, and low-cost EEG-based seizure detection system. By successfully integrating BioAmp hardware, Arduino-based filtering, and CNN classification, the system provides accurate detection and real-time visualization of neurological events.

The modular design allows for future enhancements, such as Bluetooth-based mobile alerts and cloud integration for remote clinical support. Ultimately, this innovation contributes to improved patient safety and makes essential neuro-monitoring more accessible to those living with epilepsy.

Contact Details : nishtha.jain2022@vitstudent.ac.in
mythili.asaithambi@vit.ac.in

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